

Hard + Very-hard violation rate (%)

50

40

30

20

10

0

Claude-Haiku 50%

Claude-Opus 25%

GPT-4o 19%

GPT-5.4 18%

DeepSeek-V3.2 12%

Qwen3-Max 5%

25% GPT-5.4

6% GPT-4o

6% DeepSeek-V3.2

2% Claude-Haiku

1% Claude-Opus

0% Qwen3-Max

Identification

(noisy single-variable)

Multivariate

(cross-variable coupling)

